LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Assessment of damage and estimation of loads, preparation of repair scheme.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Assessment of damage and estimation of loads, preparation of repair scheme. Material specification, special tools required for carrying out repairs
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of damage assessment techniques, load estimation calculations, and repair scheme preparation using standard tools and material specifications under instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary steps in assessing aircraft structural damage?
(b) How do you determine the load type acting on a damaged structural element?
(c) Why is material specification critical before initiating a repair scheme?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Visual inspection for damage types • Non-destructive testing methods • Load estimation principles • Material specification selection • Special tools identification	L & Ds	PPT, IB	15 min
(b)	• Damage assessment procedures • Load calculation formulas • Repair scheme formulation • Tool selection criteria • Material compatibility checks	L & Ds	PPT, PDF	20 min
(c)	• Review of repair documentation • Safety precautions for repairs • Final scheme validation • Tool maintenance protocols • Material storage guidelines	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between load estimation and repair scheme design.
(b) List three special tools required for carrying out structural repairs.
(c) Describe the importance of material specification in damage repair.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from damage assessment to repair scheme preparation for the next lesson plan (LP 02).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min